



NORTH SHORE RADIO CLUB

Flex-Cadre Radio Access

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Mission Statement

Thanks to the generosity of Burt Krain, our group has deployed a network of Flex Radios and Maestro units for the benefit of the amateur radio community. Our mission is to bring together a dedicated cadre of hams to collaborate on learning and mastering Flex Radio technology and its related software suites.

We are committed to fostering a welcoming, collaborative environment where members can learn from one another, share knowledge, and enjoy the experience of operating state-of-the-art software-defined radios remotely. Above all, we are committed to ensuring that every member has a great experience.

Our growing network currently includes five Flex Radio transceivers across four locations: two in Mundelein (EN52xg), and one each in Northfield (EN62cc), Northbrook (EN62cd), and Skokie (EN62db). As we work to expand access and participation, we are actively seeking additional hosts with versatile antenna systems and strong internet service to join our network.

File Repository

The files for this project are stored in four Github repositories:

WEB Site code - <https://github.com/mboroff/nsrc-flex-scheduler>

Node-red Flows - <https://github.com/mboroff/src-flx-scheduler-node-red-project>

VPN Creation Files - <https://github.com/mboroff/VPN-Creation-Files>

Documentation - <https://github.com/mboroff/nsrc-scheduler-documentation/tree/main>

A copy of this document is stored in the Documentation Files repository.

VPN Architecture and Access

Each of the Flex radio locations has a Raspberry Pi used to monitor for lightening strikes and control the power on/off and in some locations disconnect the antennas. These Pis also run a VPN. Marty Boroff - WD9GYM manages these Pis and user access to the VPN. The Pis are named:

- MundeleinVPN
- NorthbrookVPN
- NorthfieldVPN
- SkokieVPN

This system is located in Mundelein. Once connected to the VPN users set their browser to:

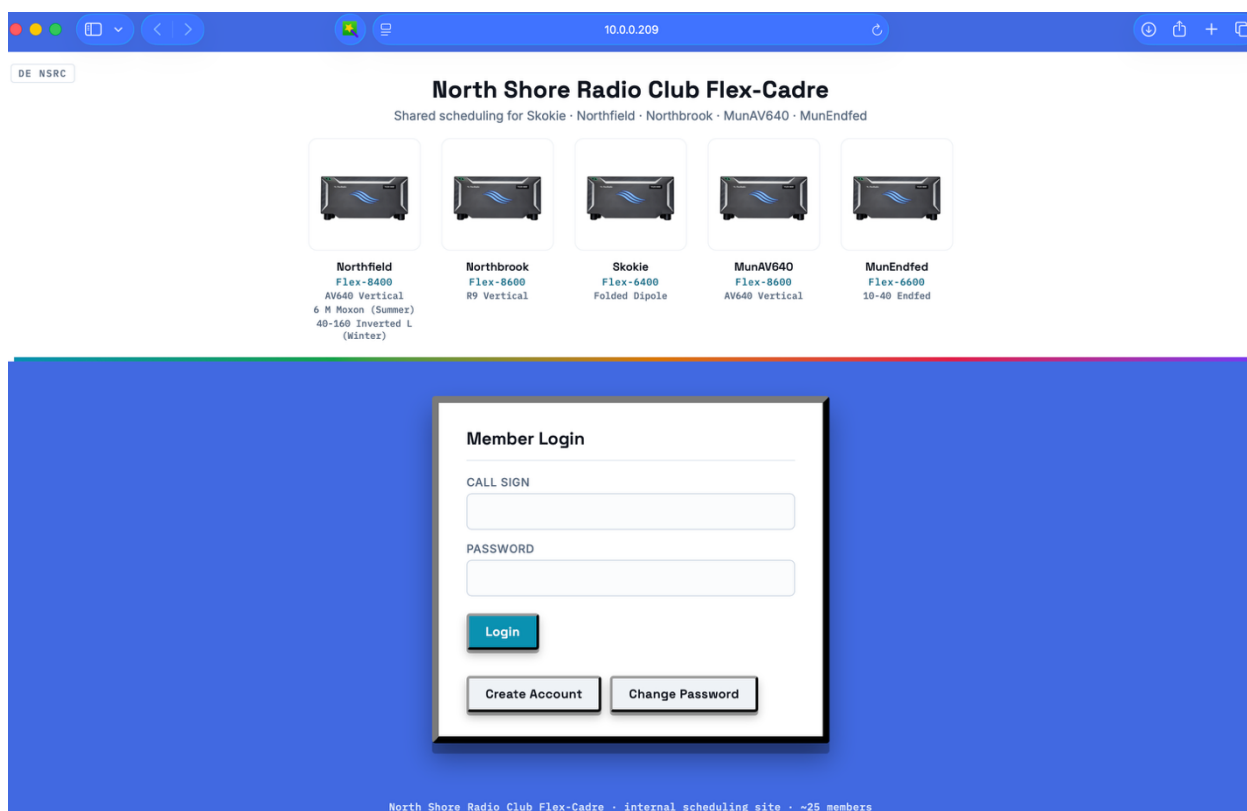
<https://mundeleinvpn.ddns.net:8090/>

This will take the user to the login page.

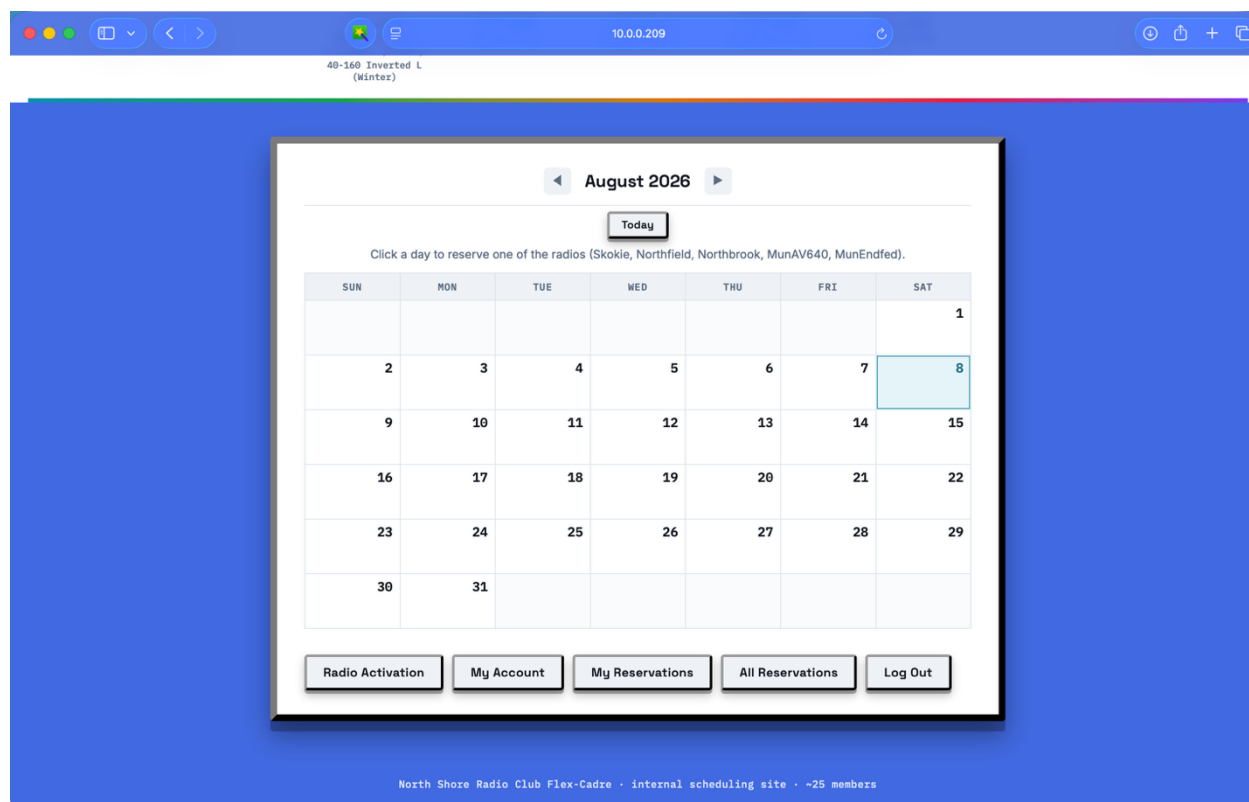
To add a user to the VPN, Marty uses the VPN server admin tool to create a user id. This is usually the user's call sign. He assigns a password and saves the information. Then on a windows computer, the Create-VPN-Package-and-letter-prompt-HTML.bat is run. This .bat file prompts for the name of the VPN connection, call sign and password. The .bat file is uploaded to Dropbox and the link is copy and pasted into the prompt. Then it generates a .bat file and three html files for use with creating an email to send to the user.

Radio Scheduling

To access the Flex radios users must login to the scheduling system after connecting to our VPN server. Once connected to the server open a browser to <http://10.0.0.209>. You will be presented with a login screen. If you do not have an account utilize the Create Account button and provide the requested information.



Once logged in, you will be presented with a calendar used to reserve time for your usage. This is not required. If you click on a date, you will be taken to the scheduling sheet for that date. If you wish to just activate a radio for connecting with SmartLink click on the Radio Activation button. Buttons are also provided to display your account information, your reservations, all reservations, and logging out.



Click on a time for a radio to reserve it. Clicking on the time again will remove the reservation.

The screenshot shows a web browser window with the address bar displaying "10.0.0.209". The page title is "Schedule a Radio - North Shore Radio Club Flex-Cadre". The main content area is titled "Saturday, August 8, 2026". Below the title, there is a instruction: "Click an open slot to reserve it under your Call Sign, or click your own reservation again to release it. Past time slots are locked." The main table has the following structure:

Time	MunAV640	MunEndfed	Northbrook	Northfield	Skokie
12:00 AM					
1:00 AM					
2:00 AM					
3:00 AM					
4:00 AM					
5:00 AM					
6:00 AM					
7:00 AM					
8:00 AM					
9:00 AM					
10:00 AM					
11:00 AM					
12:00 PM					
1:00 PM					
2:00 PM		KK9H			
3:00 PM					
4:00 PM					
5:00 PM					
6:00 PM					
7:00 PM					
8:00 PM					
9:00 PM					
10:00 PM					
11:00 PM					

At the bottom of the page, there are five buttons: "Back to Calendar", "Today", "My Reservations", "All Reservations", and "Log Out".

Clicking on My Account on the Calendar page presents this display. The user can modify any field and save it.

My Account

Created: 2026-08-08 13:29:58
Last Updated: 2026-08-09 12:45:40
Last Login: 2026-08-09 12:44:57

CALL SIGN

FIRST NAME

LAST NAME

EMAIL ADDRESS

STREET ADDRESS

CITY

STATE

ZIP CODE

PHONE NUMBER

NEW PASSWORD (LEAVE BLANK TO KEEP CURRENT)

CONFIRM NEW PASSWORD

CURRENT PASSWORD (REQUIRED TO SAVE ANY CHANGE)

Update Account

Return to Calendar

Clicking on My Reservations displays this page.

10.0.0.209

My Reservations - North Shore Radio Club Flex-Cadre

KK9H - Callsign Lookup by QRZ Ham Radio

DE NSRC

North Shore Radio Club Flex-Cadre

Shared scheduling for Skokie · Northfield · Northbrook · MunAV640 · MunEndfed



Northfield
Flex-8400
AV640 Vertical
6 M Moxon (Summer)
40-160 Inverted L
(Winter)



Northbrook
Flex-8600
R9 Vertical



Skokie
Flex-6400
Folded Dipole



MunAV640
Flex-8600
AV640 Vertical



MunEndfed
Flex-6600
10-40 Endfed

My Reservations

Every radio reservation booked under KK9H.

DATE	TIME	RADIO
Sat, Aug 8, 2026	2:00 PM - 3:00 PM	MunEndfed

[Back to Calendar](#)[All Reservations](#)[Log Out](#)

North Shore Radio Club Flex-Cadre · internal scheduling site · ~25 members

Clicking on All Reservations displays this page.

10.0.0.209

All Reservations - North Shore Radio Club Flex-Cadre

KK9H - Callsign Lookup by QRZ Ham Radio

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North Shore Radio Club Flex-Cadre

Shared scheduling for Skokie · Northfield · Northbrook · MunAV640 · MunEndfed



Northfield
Flex-8400
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Northbrook
Flex-8600
R9 Vertical



Skokie
Flex-6400
Folded Dipole



MunAV640
Flex-8600
AV640 Vertical



MunEndfed
Flex-6600
10-40 Endfed

All Reservations

Every radio reservation booked by anyone in the group.

DATE	TIME	RADIO	CALL SIGN
Sat, Aug 1, 2026	6:00 AM - 7:00 AM	Northfield	WD9GYM
Sat, Aug 1, 2026	5:00 PM - 6:00 PM	MunAV640	WD9GYM
Sat, Aug 1, 2026	6:00 PM - 7:00 PM	Northbrook	WD9GYM
Sun, Aug 2, 2026	8:00 AM - 9:00 AM	MunAV640	WD9GYM
Sun, Aug 2, 2026	11:00 AM - 12:00 PM	MunAV640	KK9H
Mon, Aug 3, 2026	10:00 AM - 11:00 AM	MunAV640	KK9H
Mon, Aug 3, 2026	11:00 AM - 12:00 PM	MunAV640	KK9H
Mon, Aug 3, 2026	12:00 PM - 1:00 PM	MunAV640	KK9H
Mon, Aug 3, 2026	1:00 PM - 2:00 PM	MunAV640	KK9H
Mon, Aug 3, 2026	2:00 PM - 3:00 PM	MunAV640	KK9H
Mon, Aug 3, 2026	3:00 PM - 4:00 PM	MunAV640	KK9H
Mon, Aug 3, 2026	4:00 PM - 5:00 PM	MunAV640	KK9H
Tue, Aug 4, 2026	4:00 AM - 5:00 AM	MunAV640	WD9GYM
Tue, Aug 4, 2026	1:00 PM - 2:00 PM	MunAV640	WD9GYM

Clicking on Radio Activation on the Calendar page takes you to this page. Click on the radio you wish to see in SmartSDR. If it's reserved by another user, nothing opens - a message shows who has it reserved instead.

Radio Activation

Click a radio to open its control dashboard in a new tab. If the current hour is already reserved by someone else, it won't open - the reservation stays protected.

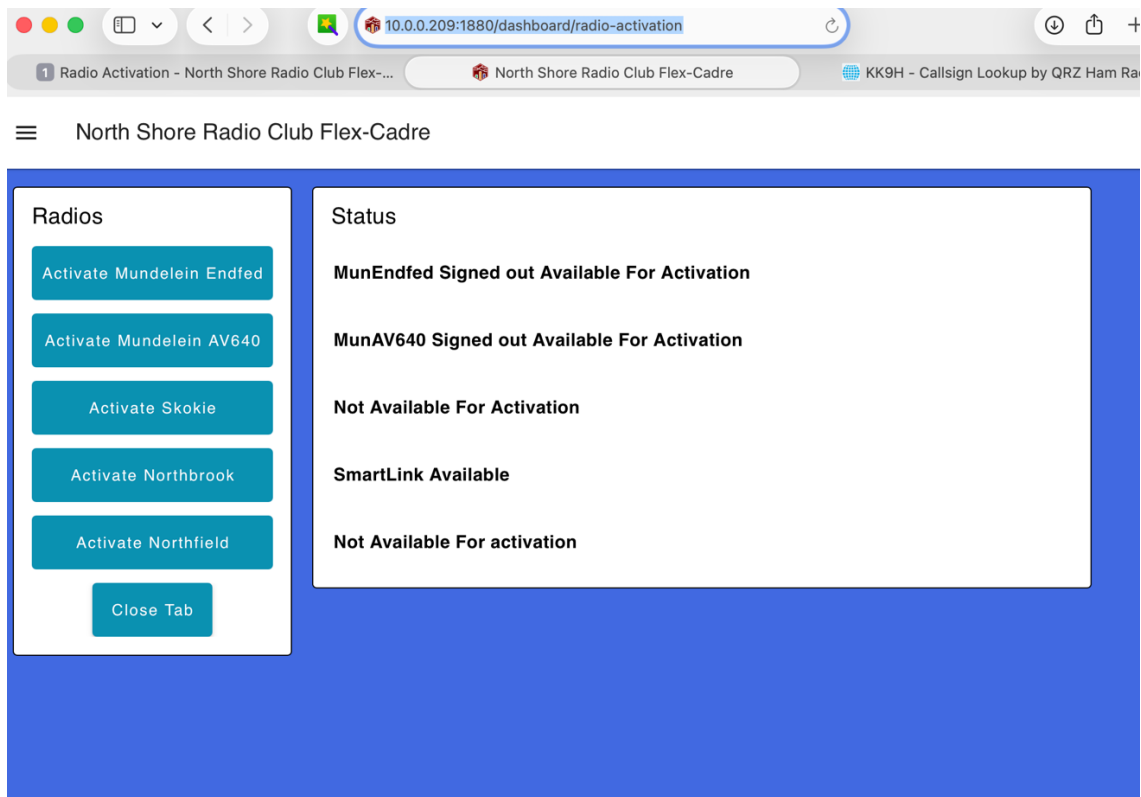
MunAV640 Flex-8600	Activate MunAV640	Radio is currently reserved by WD9GYM
MunEndfed Flex-6600	Activate MunEndfed	Available
Northbrook Flex-8600	Activate Northbrook	Available
Northfield Flex-8400	Activate Northfield	Available
Skokie Flex-6400	Activate Skokie	Available
All	Open All Radios	

[Back to Calendar](#)[Log Out](#)

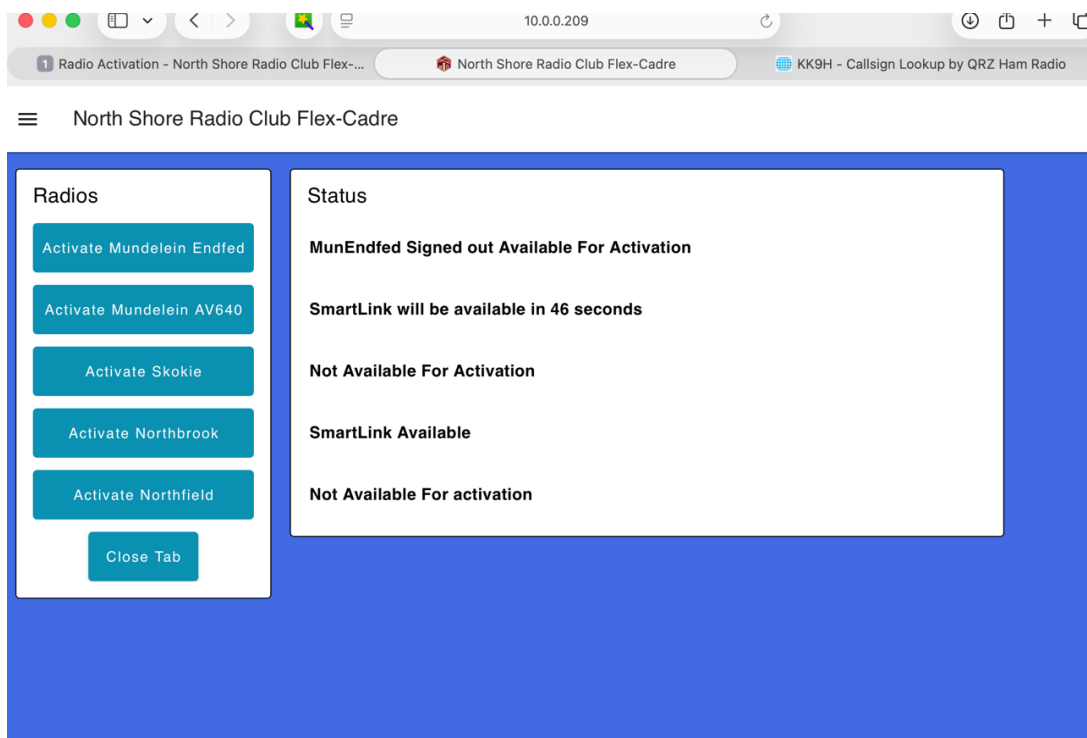
North Shore Radio Club Flex-Cadre · internal scheduling site · ~25 members

Note: the screens on the following pages belong to the club's separate Node-RED radio control system, which opens in a new browser tab when a radio is available. They are shown here for reference and are not part of the scheduling site itself.

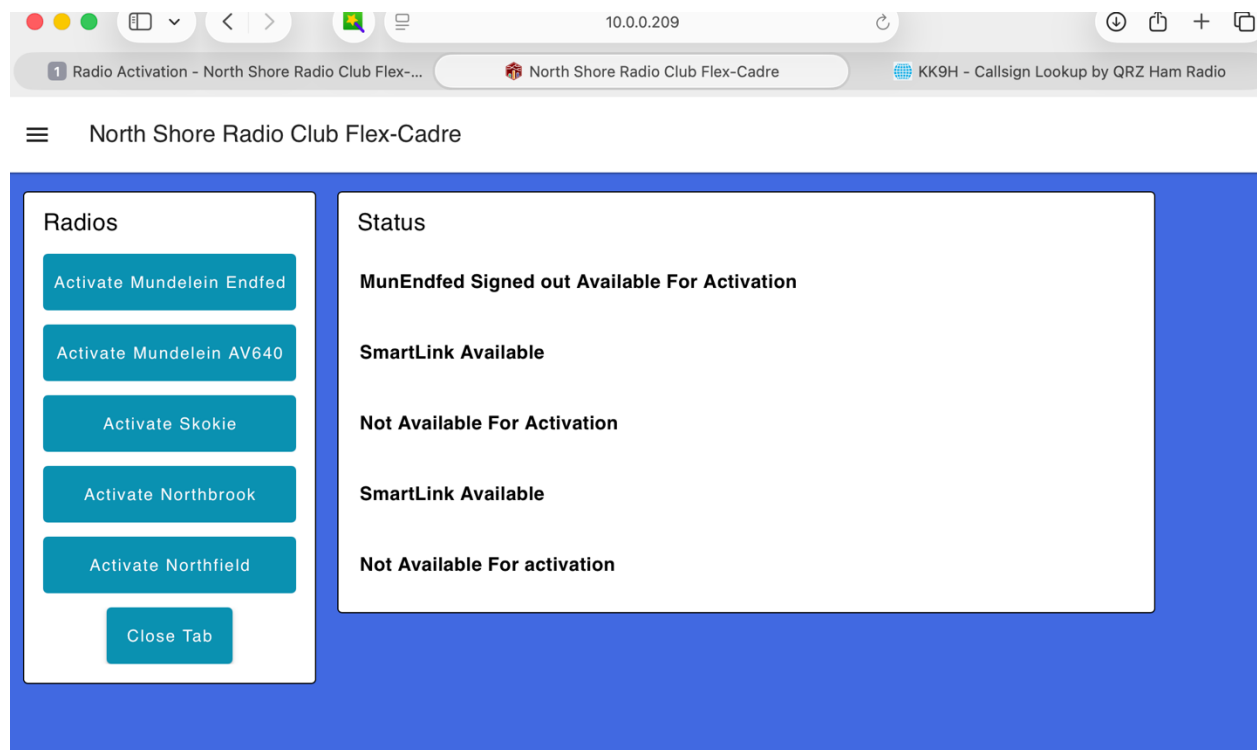
If a radio is available, you will be taken to this page for all radios or a reduced version when a specific radio is selected.



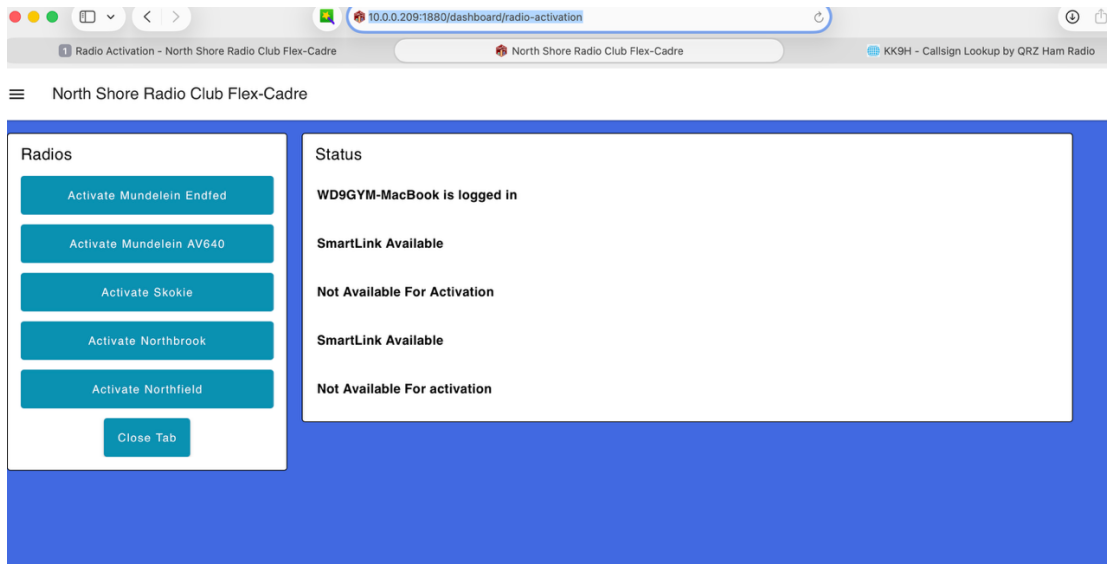
Clicking on an Activate radio button will display a count down until the radio will be available to SmartLink.



Once you see the SmartLink is available close the browser or the tab and then bring up SmartSDR application.



If a radio is in use, it is noted here.



Administering the System

When an admin logs into the system the Calendar page has an admin button located on the bottom of the page.

Scheduling - North Shore Radio Club Flex-Cadre

KK9H - Callsign Lookup by QRZ Ham Radio



Northfield
Flex-8400
AV640 Vertical
6 M Moxon (Summer)
40-160 Inverted L
(Winter)



Northbrook
Flex-8600
R9 Vertical



Skokie
Flex-6400
Folded Dipole



MunAV640
Flex-8600
AV640 Vertical



MunEndfed
Flex-6600
10-40 Endfed

◀ August 2026 ▶

Today

Click a day to reserve one of the radios (Skokie, Northfield, Northbrook, MunAV640, MunEndfed).

SUN	MON	TUE	WED	THU	FRI	SAT
						1
2	3	4	5	6	7	8
9	10	11	12	13	14	15
16	17	18	19	20	21	22
23	24	25	26	27	28	29
30	31					

Radio Activation

My Account

My Reservations

All Reservations

Admin

Log Out

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Clicking on the admin button takes you to this page.

An admin can grant or revoke admin privileges to any account, update any user information or delete an account. If the account has forgotten their password, the admin can reset it, then must let the member know the new password directly, as the system does not send this automatically.

An admin can also delete old reservations to free up space on the server.

Admin — Manage Accounts

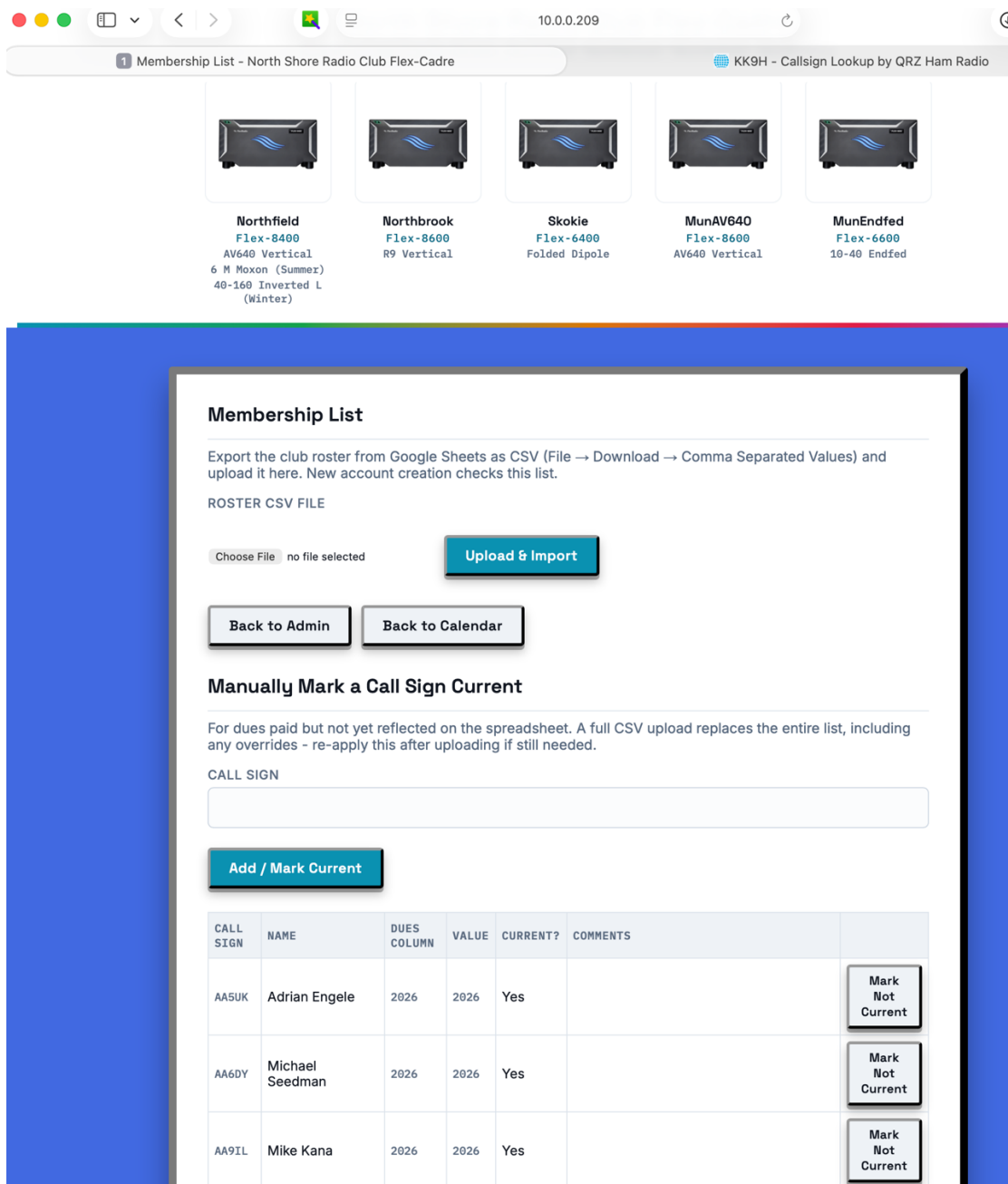
Logged in as WD9GYM. Edit any field and click Update, or click Delete to remove an account. Leave New Password blank to keep the current password.

CALL SIGN	EMAIL	NEW PASSWORD	CREATED	UPDATED	LAST LOGIN	ADMIN?	ACTIONS
KK9H	fake@addr.co	(leave blank)	2026-08-08 13:29:58	2026-08-09 12:45:40	2026-08-09 13:24:52	☐	Grant Admin Update Delete
	First Don	Last Whiteman	Phone 555-555-5555				
	Street 123 Main	City Northfield	State IL				
	Zip 60093						
WD9GYM	martin.borof	(leave blank)	2026-08-01 20:49:04	2026-08-07 22:15:27	2026-08-09 13:46:10	☒ Super	Update Delete
	First Marty	Last Boroff	Phone (847) 932-9854				
	Street 1604 Surridge Ct.	City Mundelein	State IL				
	Zip 60060						

North Shore Radio Club Flex-Cadre · internal scheduling site · ~25 members

Selecting Membership displays this page. Use this page to upload an updated list of members of the North Shore Radio Club. A list can be downloaded from the club's web site ns9rc.org. Export it to a .csv file, and remove any excess lines of information at the bottom before uploading. Select the file and then click on upload & import. This will replace the previous list.

Sometimes the member has paid their dues, but the update to the list has not been done, in which case an admin can use the Add / Mark Current option on this page to allow the user to create an account. When an account is created it is checked against this list.



Membership List

Export the club roster from Google Sheets as CSV (File → Download → Comma Separated Values) and upload it here. New account creation checks this list.

ROSTER CSV FILE

Choose File no file selected **Upload & Import**

Back to Admin **Back to Calendar**

Manually Mark a Call Sign Current

For dues paid but not yet reflected on the spreadsheet. A full CSV upload replaces the entire list, including any overrides - re-apply this after uploading if still needed.

CALL SIGN

Add / Mark Current

CALL SIGN	NAME	DUES COLUMN	VALUE	CURRENT?	COMMENTS	
AA5UK	Adrian Engele	2026	2026	Yes		Mark Not Current
AA6DY	Michael Seedman	2026	2026	Yes		Mark Not Current
AA9IL	Mike Kana	2026	2026	Yes		Mark Not Current